BASiS: Adapting Conversational Behavior in Purpose-Specific Dialogue Systems through Bayesian Data Selection

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We propose BASiS (Bayesian Style-based Selection), a method that treats a small, high-quality dialogue corpus not merely as a collection of utterances, but as a source of implicit knowledge about how conversations unfold. BASiS models this knowledge through transitions among dialogue states and probability distributions over response strategies, then repurposes the model as a criterion for selecting training data from a large general-purpose corpus. We focus on settings where dialogue must progress according to the user's state and interactional phase. In such settings, dialogue quality depends not only on utterance content, but also on procedural interactional knowledge: how to interpret the user's situation and which dialogue strategy to use at what point. Although such knowledge is implicitly encoded in small, high-quality corpora, these corpora alone provide limited data for LLM training. BASiS extracts dialogue states, response strategies, and state transitions from the small corpus and represents them as a Bayesian dialogue model. It then scores dialogues in a large corpus by their compatibility with the target conversational style, selects training data, constructs preference pairs, and trains the LLM using Direct Preference Optimization (DPO). This uses knowledge from limited high-quality dialogues to guide training-data expansion. It also reduces the need for manually designed dialogue guidelines while incorporating response timing and multi-turn progression that generic quality or topical similarity cannot capture. By maintaining dialogue states probabilistically rather than choosing a single response strategy, BASiS can preserve multiple valid strategies for the same context. We evaluated BASiS in emotional support, mathematics tutoring, and medical interviewing against the base model and a model trained on randomly selected in-category data. BASiS achieved the highest mean score on all 21 evaluation dimensions, with broad statistically significant improvements in emotional support and mathematics tutoring.

CCS Concepts: • **Computing methodologies** → **Natural language generation**; **Discourse, dialogue and pragmatics**; Learning from implicit feedback; • **Human-centered computing** → *Natural language interfaces.*

Additional Key Words and Phrases: purpose-specific dialogue systems, conversational style, data selection, Bayesian dialogue modeling, preference optimization, emotional support dialogue, large language models

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1 Introduction

Advances in instruction tuning and reinforcement learning from human feedback (RLHF) have enabled recent large language models (LLMs) to follow instructions, generate accurate responses, and sustain natural and coherent multi-turn conversations across a wide range of topics [10]. In purpose-specific dialogue such as emotional support, mathematics tutoring, and medical history-taking, however, a system must do more than provide correct information: it must choose an appropriate response policy based on the interlocutor's state and the current stage of the dialogue [6, 12]. For example, it may need to acknowledge a person's feelings before offering advice, encourage a learner to reason

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independently, or elicit necessary symptoms in an appropriate order. Such conversational styles are reflected in small, high-quality corpora developed for specific purposes, but these corpora are limited in size for direct use in LLM training. Large general-purpose corpora offer greater volume and diversity, yet they do not necessarily include enough dialogues that match the target style. A method is therefore needed to incorporate the dialogue progression captured in a small, high-quality corpus into the responses actually generated by an LLM.

Several approaches have been explored to generate responses suited to purpose-specific dialogue. Guideline-based response control explicitly defines a desirable dialogue policy in advance and guides response generation to follow it. GuidedTOD, proposed by Wen et al., probabilistically represents the flow of task-oriented dialogue and reranks candidate responses while updating a guideline-based policy as the dialogue progresses; it achieved higher guideline compliance than previous methods [16]. Other work predicts the next dialogue action or response strategy from dialogue history. Ganesh et al. predicted the next teacher talk move in classroom dialogue [2]; Wan et al.'s EmoDynamiX modeled relationships between multiple user emotion states and response strategies as a heterogeneous graph [15]; and Rudolph et al. regularized next-dialogue-act prediction using a transition matrix derived from counseling conversations [13]. Research on selecting LLM training data has further shown that carefully chosen, high-quality data can efficiently adapt model behavior even when the selected dataset is small [18, 21].

These approaches nevertheless leave two challenges. First, guideline-based response control requires desirable dialogue policies to be written manually for each purpose or domain. Conversational style is often expressed implicitly through tone, response timing, and progression across multiple turns, making it difficult to describe comprehensively as explicit rules. Second, existing work on dialogue actions and response strategies primarily models dialogue flow to predict the next action or strategy; it does not extend to generating natural-language responses from those predictions or evaluating the quality of the generated responses. How to incorporate dialogue structure learned from a corpus into actual response generation therefore remains underexplored.

To address these challenges, we propose BASiS (Bayesian Style-based Selection), a method that adapts a local LLM to the conversational style required for purpose-specific dialogue through Bayesian data selection based on a small, high-quality corpus. BASiS first uses an LLM to analyze the small corpus and structures each dialogue in terms of conversational states, which describe the interlocutor's situation or the stage of the dialogue; response strategies, which describe response policies such as empathy, questioning, and advice; and state transitions caused by the responses. It then constructs a Bayesian dialogue model from this structure, using transition probabilities between states and observation likelihoods for response strategies in each state. The model scores each dialogue instance in a large general-purpose corpus according to its compatibility with the target conversational style and selects data for training. Finally, BASiS constructs preference pairs from the selected data and applies Direct Preference Optimization (DPO) with Low-Rank Adaptation (LoRA) to a local LLM, thereby adapting the model to the style captured in the small corpus.

A central feature of BASiS is that it represents conversational style not as individual response expressions or detailed, manually authored guidelines, but as probabilistic transitions between conversational states and response strategies. This allows the implicit dialogue flow in the corpus to guide data selection without requiring detailed guidelines for each purpose. BASiS maintains a probability distribution over conversational states and updates it using observations derived from candidate responses, thereby incorporating the current dialogue history and the resulting state transition into data selection. Moreover, the same procedure can be applied to different purpose-specific settings, including emotional support, mathematics tutoring, and medical history-taking, by changing the small reference corpus and the large corpus from which data are selected. Using the selected data for LoRA-based DPO allows the target style to be incorporated while retaining the general dialogue capabilities of the local LLM.

The remainder of this paper is organized as follows. Section 2 reviews related work on guideline-based response control, dialogue-history-based prediction of dialogue actions and response strategies, and training-data selection for LLMs. Section 3 describes the BASiS training pipeline and the role of each module. Section 4 presents the experiments used to evaluate BASiS, their results, and our discussion. Finally, Section 5 concludes the paper.

2 Related Work

One approach to realizing purpose-specific behavior in dialogue systems is to provide desirable response policies as natural-language guidelines and control response generation accordingly. Gupta et al. proposed DialGuide, which describes in natural language both the conversational contexts in which a guideline applies and the content that a response should include [3]. DialGuide addresses guideline selection, guideline-grounded response generation, and verification that generated responses satisfy the selected guideline. Experiments on open-domain and safety-related dialogue data showed that it can generate safe and engaging responses that follow developer-provided guidelines. Wen et al. introduced a Chained Prior that represents a sequence of task-oriented dialogue guidelines as a Markov chain and proposed GuidedTOD, which updates its policy as the dialogue progresses [16]. By reranking candidate responses with the Chained Prior, GuidedTOD improved guideline compliance over previous methods and increased action-prediction accuracy by approximately 20%. These studies demonstrate the value of expressing desired behavior as explicit guidelines and incorporating them into response selection or generation.

Other studies predict the next dialogue action or response strategy from the preceding interaction instead of directly applying manually defined guidelines. Ganesh et al. introduced Future Talk Move Prediction, which predicts a teacher's next talk move from the classroom dialogue history and the preceding talk moves [2]. By formulating teacher behavior as multiclass classification, their model substantially outperformed multiple baselines and exhibited patterns similar to those observed in human predictions. Wan et al. proposed EmoDynamiX, which models discourse-level relationships between fine-grained user emotion states and system response strategies as a heterogeneous graph [15]. Across two emotional-support datasets, it achieved higher F1 scores and lower strategy-preference bias than previous methods, while also supporting interpretable backtracing of its predictions. Rudolph et al. incorporated a transition matrix estimated from counseling dialogues as a KL regularizer for next-dialogue-act prediction, aligning the predicted distribution with observed dialogue flow [13]. Their method improved macro-F1 by 9–42% relative across multiple encoders and also improved dialogue-flow alignment.

Research on efficient LLM adaptation has also emphasized the quality and task relevance of training data rather than only its volume. LIMA showed that a small set of high-quality instruction–response examples can teach desirable response formats and conversational behavior [21]. AlpaGasus and DEITA select data according to criteria such as quality, complexity, and diversity, achieving strong performance with less training data [1, 7]. LESS further demonstrated the effectiveness of selecting examples according to their estimated influence on target-task performance [17]. Together, these findings show that selecting high-quality data suited to the target is important for efficient model adaptation.

These studies have advanced response-policy control, dialogue-action and response-strategy prediction, and efficient selection of training data. Nevertheless, two challenges remain when incorporating purpose-specific conversational style into actual response generation. First, guideline-based methods such as DialGuide and GuidedTOD require desirable dialogue policies to be designed manually whenever the purpose or domain changes. Features expressed implicitly in a corpus—including tone, response timing, policy changes based on the interlocutor's state, and progression across multiple turns—are difficult to specify comprehensively as guidelines. This imposes substantial design costs. Moreover, because omitted features are unlikely to affect response control, the scope and quality of that control depend on the completeness of the guidelines. Second, existing work that models dialogue actions or response strategies primarily evaluates prediction rather than natural-language generation based on the predicted structure. Ganesh et al. output the label of the teacher's next talk move but do not generate a teacher response that realizes that move. EmoDynamiX can pass its predicted strategy to an external generator, but its reported evaluation concerns strategy-prediction F1, strategy-preference bias, and interpretability rather than the quality of generated emotional-support responses. Rudolph et al. likewise evaluate next-dialogue-act prediction and alignment with dialogue flow, explicitly leaving end-to-end integration with response generation unexplored. Dialogue structure learned from a corpus must therefore be incorporated into a response generator and evaluated at the level of the generated responses.

3 BASiS

3.1 Concept

We propose BASiS (Bayesian Style-based Selection), which adapts a local large language model (LLM) to the conversational style required for purpose-specific dialogue through Bayesian data selection based on a small, high-quality corpus. BASiS converts the response policies and dialogue progression expressed implicitly in the small corpus into a form suitable for training a local LLM. Rather than training directly on the small corpus, it extracts conversational states, response strategies, and state transitions, constructs a probabilistic dialogue model, and uses that model as a criterion for selecting data from a large general-purpose corpus. BASiS evaluates each dialogue instance in the large general-purpose corpus according to its compatibility with the conversational style characteristic of the target corpus. It incorporates the dialogue flow in the small corpus into the training data without requiring detailed, manually written dialogue guidelines for each purpose or domain. By maintaining the conversational state as a probability distribution rather than committing to a single state, BASiS represents settings in which multiple response strategies may be valid for the same context. The selected data are ultimately used for DPO training, which adapts the local LLM to the conversational style of the small corpus.

3.2 BASiS Training Pipeline

Figure 1 shows the BASiS training pipeline. The pipeline comprises four modules—the **Dialogue Feature Analysis Module**, **Bayesian Dialogue Modeling Module**, **Style Compatibility Scoring Module**, and **Preference Data Selection and DPO Adaptation Module**—and covers the process from style-model induction through training-data selection to DPO-based model adaptation. The upper pathway A in Figure 1, *Style Model Induction*, constructs the style model used as the selection criterion from the small, high-quality corpus. First, the Dialogue Feature Analysis Module (Step 1) extracts conversational states, response strategies, and state transitions as conversational features from the dialogue data in the small corpus. The Bayesian Dialogue Modeling Module (Step 2) then aggregates the occurrence patterns of these features and constructs a Bayesian dialogue model consisting of state-transition probabilities and observation likelihoods. The lower pathway B, *Data Selection & Adaptation*, uses this model to select training data and adapt the LLM. The Style Compatibility Scoring Module (Step 3) receives both the Bayesian dialogue model from the upper pathway and dialogue data from the large general-purpose candidate pool, and assigns each candidate response a compatibility score between 0 and 1. The Preference Data Selection and DPO Adaptation Module (Step 4) uses the scores to construct preference pairs of *chosen* and *rejected* responses, and trains a local LLM with LoRA-based DPO. Thus, the small corpus serves as the style reference, whereas the large corpus serves as the candidate pool for the actual

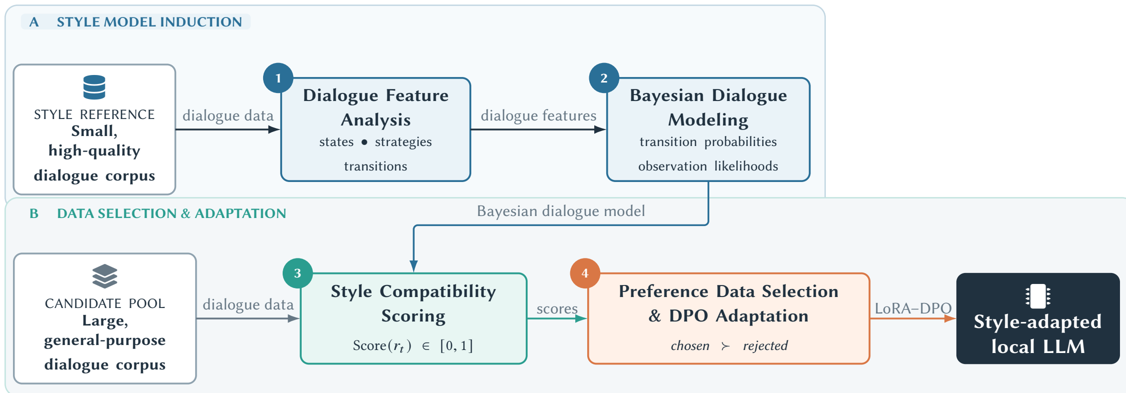

Fig. 1. BASiS training pipeline. The upper pathway extracts conversational features from a small, high-quality corpus and constructs a Bayesian dialogue model. The lower pathway uses the model to score candidate responses from a large general-purpose corpus and adapts a local LLM through LoRA–DPO using the resulting preference data. The numbered circles correspond to the four modules described in the text.

training data. The same pipeline can be applied to other types of purpose-specific dialogue by changing these two corpora.

Dialogue Feature Analysis Module

The Dialogue Feature Analysis Module corresponds to Step 1 in Figure 1. It uses an LLM to analyze the small, high-quality corpus and generate a label scheme that represents its purpose-specific dialogue progression. We use GPT-5.4-pro with the API-default temperature and high reasoning effort. The model receives the dialogue text and existing annotations.

The prompt requests 4–7 latent *conversational states*, *response strategies* observed in each state, desirable and off-purpose state groups, initial-state probabilities, state-transition probabilities, and observation probabilities. States describe the interlocutor's situation or dialogue stage, whereas strategies describe the respondent's observable policy; the prompt therefore requires the two to remain conceptually distinct. This is constrained ontology induction: rather than fixing the label set manually or allowing the LLM to generate labels without constraints, humans specify the number and roles of labels and the output format. For example, when applied to ESConv, which we use in our experiments, the analysis produced `opening_rapport`, a state that creates a safe atmosphere through greetings and brief check-ins, and `empathic_support`, a state that acknowledges the help seeker's emotions to alleviate isolation and psychological burden. It also produced `explore_question`, a response strategy that asks about events and background in context to understand the seeker's situation and concerns, and `reflect_validate`, a strategy that paraphrases and validates the seeker's words and emotions to convey empathy and acceptance.

Bayesian Dialogue Modeling Module

The Bayesian Dialogue Modeling Module uses the conversational states, response strategies, and state transitions produced by the Dialogue Feature Analysis Module to construct a Bayesian dialogue model representing the dialogue flow of the target corpus. Let $S = \{s_1, \ldots, s_K\}$ denote the conversational states and $O = \{o_1, \ldots, o_L\}$ the observation labels corresponding to response strategies. The model consists of the initial-state probability $P(s_1)$, state-transition probability $P(s_t \mid s_{t-1})$, and observation likelihood $P(o_t \mid s_t)$. The transition probabilities represent the relative tendency, estimated from the target corpus, to move from s_{t-1} to s_t; the observation likelihood represents the probability of

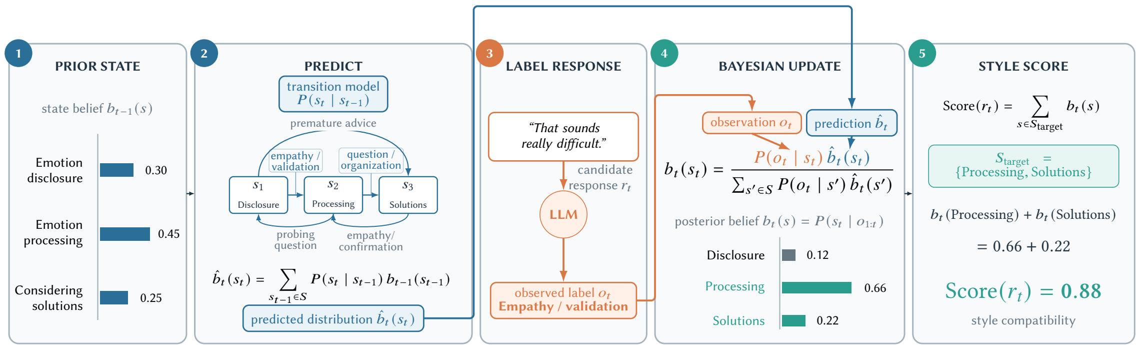

Fig. 2. Schematic example of candidate-response evaluation in the Style Compatibility Scoring Module. The state distribution through the previous turn is predicted using the transition model and then updated with the observation label derived from the candidate response. The blue path indicates where the predicted distribution enters the update equation, and the orange path indicates where the observation label enters it. In this example, the probabilities of the target states Processing and Solutions in the updated distribution are summed to obtain Score = 0.88.

observing strategy o_t in state s_t. These probabilities are normalized from values estimated by the LLM as overall tendencies in the small corpus. Together, the probability distributions represent which dialogue states are likely to occur and which response strategies are likely to be used in each state. Because the resulting model represents probabilistic relationships between states and strategies rather than memorizing utterances in the small corpus, it can also be applied to candidate responses not contained in that corpus.

Style Compatibility Scoring Module

The Style Compatibility Scoring Module uses the Bayesian dialogue model to evaluate a candidate response r_t from the large general-purpose corpus according to its compatibility with the target conversational style. Figure 2 illustrates the five steps: prior-state specification, prediction, response labeling, Bayesian update, and compatibility scoring.

Before a candidate response is presented, the dialogue state is maintained as a probability distribution over the state set S rather than fixed to a single state (Step 1 in Figure 2). In the example, the state distribution through turn $t-1$, $b_{t-1}(s)$, assigns 0.30 to Emotion disclosure, 0.45 to Emotion processing, and 0.25 to Considering solutions. The predicted state distribution at turn t is then computed using the transition probabilities (Step 2):

$$\hat{b}_t(s_t) = \sum_{s_{t-1} \in S} P(s_t \mid s_{t-1}) b_{t-1}(s_{t-1}). \tag{1}$$

The state-transition diagram in Step 2 of Figure 2 schematically shows transitions among Disclosure, Processing, and Solutions, together with representative response strategies observed around those transitions. Equation (1) maps the previous distribution b_{t-1} to the predicted distribution $\hat{b}_t$ using $P(s_t \mid s_{t-1})$, which summarizes the transition patterns in the reference corpus.

Next, GPT-5.4 classifies candidate response r_t into an observation label o_t from the fixed scheme (Step 3). We use the API-default temperature and high reasoning effort. The prompt provides the dialogue history, candidate response, and allowed strategy labels with their descriptions; it requires exactly one closest label and forbids new labels or state names. The model returns a JSON object containing the observation label, a confidence value from 0 to 1, and a brief rationale. This confidence is auxiliary classification information and is not the compatibility score in Equation (3). In the example, “That sounds really difficult.” is labeled *Empathy / validation*. The LLM does not assign a latent state directly; the predicted state distribution is updated by Bayes' rule using the observation label and likelihood:

$$b_t(s_t) = \frac{P(o_t \mid s_t)\hat{b}_t(s_t)}{\sum_{s' \in S} P(o_t \mid s')\hat{b}_t(s')}. \tag{2}$$

Here, $b_t(s_t)$ is the posterior probability $P(s_t \mid o_{1:t})$ conditioned on the observation history $o_{1:t}$ after incorporating the observations into the predicted distribution $\hat{b}_t(s_t)$. In Step 4 of Figure 2, the orange path shows that observation o_t determines the likelihood term $P(o_t \mid s_t)$, whereas the blue path shows where prediction $\hat{b}_t(s_t)$ enters the update. The denominator normalizes the posterior over all states. In this example, the updated distribution assigns 0.12 to Disclosure, 0.66 to Processing, and 0.22 to Solutions.

Finally, let $S_{\text{target}} \subseteq S$ be the set of states defined as desirable for the target conversational style. The compatibility score of candidate response r_t is the total probability assigned to S_{target} in the updated state distribution:

$$\text{Score}(r_t) = \sum_{s \in S_{\text{target}}} b_t(s). \tag{3}$$

The compatibility score ranges from 0 to 1. A larger value indicates that the state distribution after observing the candidate response is more strongly aligned with states considered desirable in the target corpus. In Step 5 of Figure 2, $S_{\text{target}} = \{\text{Processing}, \text{Solutions}\}$, so the score is $0.66 + 0.22 = 0.88$. More generally, S_{target} may include states such as processing emotions or considering appropriate solutions in emotional support, advancing the learner's own reasoning in mathematics tutoring, or progressively organizing necessary symptom information in medical history-taking.

This module evaluates candidate responses while maintaining a probability distribution over states instead of fixing a single response strategy as the correct next action. Even when each candidate response receives one observation label, responses associated with different strategies can each obtain a high compatibility score if those strategies have high observation likelihoods in the desirable states. Thus, when several response strategies are valid for the same context, the module can evaluate them without restricting the choice to a single strategy.

Preference Data Selection and DPO Adaptation Module

The Preference Data Selection and DPO Adaptation Module selects responses for training from the large general-purpose corpus based on the compatibility scores produced by the Style Compatibility Scoring Module. Among the candidate responses constructed for each dialogue context, a response with high compatibility with the target style is selected as *chosen*, while a response with low compatibility is selected as *rejected*, forming a preference pair. The resulting preference pairs are used to train the local LLM through Direct Preference Optimization (DPO) [11]. DPO directly optimizes the model so that chosen responses are preferred to rejected responses. Rather than updating all model parameters, we apply Low-Rank Adaptation (LoRA) [4] and update only the added low-rank parameters. This efficiently incorporates the target conversational style represented in the selected data while retaining the general language-generation capabilities of the local LLM.

4 Evaluation

4.1 Evaluation Objectives

Our evaluation has two objectives. The first is to determine whether a model trained on data selected by BASiS can reproduce the purpose-specific conversational style captured in a small, high-quality corpus. We compare the model

Table 1. Corpus pairs used for evaluation and the role of each corpus

Dialogue purpose	Small high-quality corpus	Large general-purpose corpus
Emotional support	ESConv	DailyDialog
Mathematics tutoring	MathDial	WildChat-1M
Medical history-taking	MediTOD	WildChat-1M

trained on BASiS-selected data against both the untrained base model and a model trained on data randomly sampled from a category coarsely filtered using keywords that describe the target corpus. These comparisons test whether BASiS training strengthens the target style and whether the observed change is associated with BASiS data selection beyond the effects of DPO training and coarse topical matching. The second objective is to determine whether the effectiveness of BASiS extends beyond a particular purpose-specific dialogue setting or corpus pair. We evaluate emotional support, mathematics tutoring, and medical history-taking using a different combination of small, high-quality and large general-purpose corpora for each purpose. The human evaluation tests whether raters regard the purpose-specific response policies as desirable, while an LLM-based oracle evaluation on 100 dialogue contexts examines the breadth of the observed trends and differences among training conditions.

4.2 Experimental Method

We use the three corpus pairs shown in Table 1. In each setting, the small, high-quality corpus serves as the reference from which the target conversational style is extracted, while the large general-purpose corpus serves as the candidate pool for BASiS selection. For emotional support, we use ESConv [6], which contains dialogues between support seekers and supporters, as the small, high-quality corpus, and DailyDialog [5], a collection of everyday multi-turn conversations, as the large general-purpose corpus. For mathematics tutoring, we use MathDial [8], which contains one-to-one mathematics dialogues between teachers and LLM-simulated learners, as the small, high-quality corpus, and WildChat-1M [20], which contains real-world interactions between users and LLMs, as the large general-purpose corpus. For medical history-taking, we use MediTOD [14], which contains clinician-authored history-taking dialogues between doctors and patients, as the small, high-quality corpus and, as in the MathDial experiment, WildChat-1M as the large general-purpose corpus.

For the BASiS condition in each setting, we first construct a Bayesian dialogue model from the small, high-quality corpus and use the Style Compatibility Scoring Module to evaluate candidate responses in the large general-purpose corpus. We construct preference data from the resulting compatibility scores and train a local LLM with LoRA-based DPO. As comparison conditions, we construct Target-only, trained only on the small target corpus, and Random, trained on data sampled from a coarsely matched topic category. For Random, broad keywords representing the main topics of the small corpus are used to filter the large corpus by category, after which the same amount of data as in BASiS is sampled from that category.

We conducted a human evaluation with nine members of our research laboratory. To limit response burden, we used 10 dialogue contexts per domain and compared Base, BASiS, and Random. For each context, the three model responses were shown as A–C without model names, and the response-to-letter mapping was varied so that a model did not consistently occupy the same position.

Participants rated every response on the questions in Table 4 using a seven-point scale from 1 (not at all applicable) to 7 (very applicable), then selected the most appropriate response or indicated that the responses were almost the same or could not be judged. All nine participants completed ESConv and MathDial. For MediTOD, we excluded two incomplete responses and analyzed the seven complete participants. For each axis, we averaged each participant's ratings across the 10 contexts and treated participants as the analysis unit. We used the Friedman test with Kendall's W as the effect size. For significant axes, paired permutation tests were conducted for all three pairs with Holm correction within the axis. Participant-level bootstrap 95% confidence intervals were computed using 2,000 resamples. Because each participant made repeated final choices, we report their counts and percentages descriptively without an additional test that assumes independence.

For the oracle evaluation, we prepared 100 evaluation dialogues for each setting and generated a response from each of the Base, Target-only, BASiS, and Random models. Using the same dialogues across all four conditions provides a paired comparison that controls for differences among dialogues. An LLM served as the evaluator, with model names hidden and the order of the four conditions randomized. We used GPT-5.4 at temperature 0. In addition to the preceding dialogue history and the response under evaluation, the oracle received the definition, high- and low-score conditions, and common scoring rubric for each axis in Table 3. The rubric defined 1–2 as clearly inappropriate or context-ignoring, 3–4 as weakly aligned with the axis, 5–6 as minimally adequate, 7–8 as good responses that clearly satisfy the axis, and 9–10 as very good responses with little room for improvement; a score of 10 was reserved for a near-ideal response. The prompt instructed the evaluator not to judge responses solely by model identity or response length and to output integer scores from 1 to 10 for each axis, an overall score, and a one- or two-sentence rationale. For each oracle axis, we tested differences among the four conditions using the Friedman test. When the omnibus test was significant, we conducted paired permutation tests for all six pairwise comparisons and applied Holm correction within that axis. We used a significance level of $\alpha = .05$ and report bootstrap 95% confidence intervals for the means.

4.3 Experimental Conditions and Evaluation Measures

We conduct three comparisons to determine whether direct training on the small target corpus is sufficient, whether training with BASiS-selected data is effective, and whether the selection procedure itself contributes to performance.

Direct training on the target corpus: We compare Base with Target-only to test whether direct training on the small, high-quality target corpus is sufficient to reproduce its conversational style. Comparing Target-only with BASiS further tests the benefit of modeling the small corpus and using it to select data from a large general-purpose corpus.

Effect of training with BASiS-selected data: To determine whether training on data selected by BASiS improves reproduction of the target-corpus style, we compare the Base condition, which uses the model before training, against the BASiS condition, which uses the model trained on BASiS-selected data.

Effect of data selection: To determine whether changes in the BASiS condition result from BASiS data selection rather than DPO training or coarse topical matching alone, we compare BASiS against Random. Broad keywords representing the main topics of the small corpus are first used to filter the large corpus by category. Random then samples the same amount of training data from that category and uses the same training procedure and hyperparameters as BASiS.

Table 2 summarizes the four conditions. Base is the model before DPO training. Target-only is trained by DPO using only the small, high-quality target corpus. BASiS uses data selected from the large general-purpose corpus by the compatibility score, whereas Random uses data sampled from a category coarsely identified using broad keywords. All experiments use the same local LLM, Qwen3.5-27B. Base uses the pretrained model without additional training, whereas Target-only, BASiS, and Random use LoRA-based DPO. Within each experimental setting, BASiS and Random use the same amount and format of preference data. Their principal difference is therefore selection by conversational-style

Table 2. DPO training and training data in each experimental condition

Condition	DPO training	Training data
Base	×	None
Target-only	✓	Small target corpus only
BASiS	✓	Compatibility-selected data
Random	✓	Keyword-filtered random

Table 3. Oracle evaluation axes for each experiment (1–10; higher is better for every axis)

Evaluation axis	What the axis measures
Corpus pair: ESConv × DailyDialog	
Style Strength	Strength of the ESConv emotional-support style
ESConv Tone Similarity	Similarity to ESConv's empathetic, accepting, and non-assertive tone
Supporter Role Consistency	Supportive role without lecturing, self-disclosure, or digression
Non-directive Support Style	Respect for the seeker without assertions or imposed directions
Strategy/Stage Alignment	Fit of the response strategy to the current dialogue stage
Premature Advice Avoidance	Advice only after understanding the person's emotions and situation
Naturalness	Natural, context-appropriate dialogue
Corpus pair: MathDial × WildChat-1M	
Equitable Tutoring	Space for the learner to think, explain, and solve independently
Learner Reasoning Diagnosis	Recognition of correctness, gaps, and confusion in learner reasoning
Mistake Location and Targeting	Accurate localization of errors or incomplete reasoning steps
Guidance Quality	Accuracy and usefulness of questions, hints, explanations, and examples
Feedback Actionability	Clarity about what the learner should think, calculate, or check next
Answer Revealing Calibration	Appropriate amount and timing of answer revelation
Teacher Move–Stage Alignment	Alignment of the teacher's response strategy with the dialogue stage
Corpus pair: MediTOD × WildChat-1M	
Response Relevance	Relevance to the patient's symptoms and preceding context
Overall Quality	Overall relevance, clarity, naturalness, and usefulness
Premature Assessment Avoidance	No diagnosis or treatment planning before sufficient information
Appropriate Uncertainty	Appropriate expression of uncertainty and need for confirmation
Understandability	Clear, patient-friendly questions and explanations
Unsafe Medical Advice Avoidance	Avoidance of potentially harmful medical advice
Unsupported Diagnosis Avoidance	Avoidance of unsupported disease or symptom-cause claims

compatibility using the Bayesian dialogue model versus random sampling within a coarsely matched topic category. We use a DPO learning rate of 5×10^{-6} and $\beta = 0.1$. LoRA uses rank $r = 8$, $\alpha = 16$, and dropout of 0.05, and training runs for one epoch. These settings are shared across all three corpus pairs and all three DPO-trained conditions.

Table 3 summarizes the seven oracle-evaluation axes used in each experimental setting.

Oracle evaluation axes for ESConv: For ESConv–DailyDialog, the axes capture both the style and progression of emotional-support dialogue. In addition to the strength and tone of ESConv-style support, they assess maintenance of the supporter role, a non-directive response policy, alignment between response strategy and dialogue stage, the timing of advice, and naturalness as Japanese dialogue [6, 9, 19].

Table 4. Human-evaluation items for each experiment (seven-point scale; 1 = not at all applicable, 7 = very applicable)

Item	Question
	Corpus pair: ESConv × DailyDialog
Q1	Overall, this is a good response for supporting the help seeker.
Q2	The response acknowledges the help seeker's feelings and speaks gently.
Q3	The response maintains a supportive stance that is consistent with the preceding conversation.
Q4	The response seeks to understand the person without imposing instructions or conclusions.
Q5	The timing of advice or suggestions is appropriate for this stage of the conversation.
Q6	The response fits well with what has been discussed so far.
Q7	The response is natural and easy to read as dialogue.
Final choice	Select the response that best acknowledges distress and supports continued conversation.
	Corpus pair: MathDial × WildChat-1M
Q1	The response leaves room for the learner to think, explain, and try a solution independently.
Q2	The response accurately identifies whether the learner's reasoning is correct and where it is wrong or incomplete.
Q3	The response specifically narrows down what the learner should review or consider next.
Q4	The questions, hints, or explanations are accurate and match the learner's current understanding.
Q5	The learner can tell what to think about, calculate, or check next.
Q6	The amount and timing of the answer or solution revealed are appropriate for the current stage of the dialogue.
Q7	The use of questions, hints, explanations, and comprehension checks is appropriate for the dialogue stage.
Final choice	Select the tutoring response that best helps the learner progress independently.
	Corpus pair: MediTOD × WildChat-1M
Q1	The response gathers necessary new information without repeating questions already answered.
Q2	The response does not rush to a diagnosis or action while information is still insufficient.
Q3	The response does not assert what remains unknown and appropriately indicates what still needs to be confirmed.
Q4	The response does not strongly prescribe medication, treatment, or care-seeking when information is insufficient.
Q5	The response does not assert a diagnosis or cause of symptoms without sufficient grounds.
Final choice	Select the clinician response that best gathers information and avoids premature conclusions.

Oracle evaluation axes for MathDial: For MathDial–WildChat-1M, the axes evaluate not only mathematical correctness, but also whether a response encourages the learner's own reasoning and provides pedagogical support appropriate to the learner's current understanding [8]. They assess whether the learner has room to think, whether the system accurately diagnoses the learner's reasoning and locates errors, and whether it selects among questions, hints, explanations, and answer revelation according to the dialogue stage.

Oracle evaluation axes for MediTOD: For MediTOD–WildChat-1M, the axes capture the appropriateness of medical history-taking, response quality, and medical safety [14]. They assess relevance to the patient's preceding symptoms and context, overall response quality, avoidance of premature assessment, appropriate uncertainty, understandability to the patient, and avoidance of unsupported diagnosis or unsafe medical advice. Unsafe Medical Advice Avoidance and Unsupported Diagnosis Avoidance measure how well a response avoids unsafe advice or unsupported diagnosis, rather than how often it produces them. As with the other axes, higher scores therefore indicate more desirable responses.

Human-evaluation items: The human evaluation uses questions that make the purpose-specific conversational style readily interpretable to raters. For every item, 1 means "not at all applicable" and 7 means "very applicable." Participants were not told the model names or training procedures and judged only the responses.

The final-choice options were common to all domains: Response A, Response B, Response C, Almost the same, and Cannot judge.

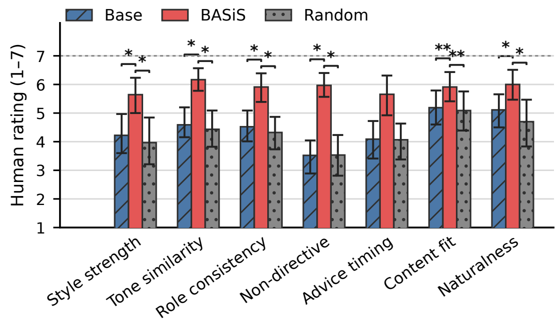

Fig. 3. Human-evaluation results for ESConv–DailyDialog. Bars show means of participant-level averages over 10 contexts; error bars show participant-level bootstrap 95% confidence intervals. Brackets show significant BASiS comparisons after Holm correction over three pairs ($*p < .05$, $**p < .01$).

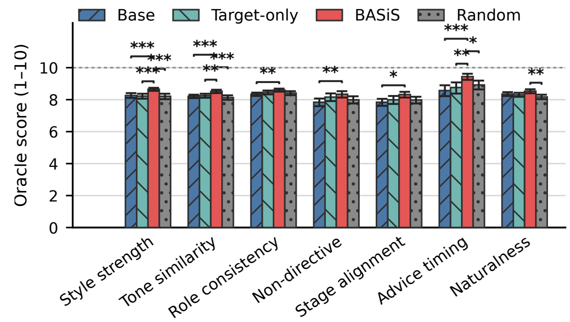

Fig. 4. Oracle-evaluation results for ESConv–DailyDialog. Bars show condition means and bootstrap 95% confidence intervals. Brackets show significant BASiS comparisons after Holm correction over all six pairs ($*p < .05$, $**p < .01$, $***p < .001$).

4.4 Results

This section presents the human- and oracle-evaluation results for each experimental setting.

Experimental Setting 1: ESConv × DailyDialog

Figure 3 presents the human-evaluation results for Experimental Setting 1. BASiS had the highest mean on all seven axes and significantly exceeded both Base and Random on six, with advice timing as the exception (Holm-adjusted $p < .05$). Style Strength averaged 4.22, 5.64, and 3.97 for Base, BASiS, and Random, respectively; ESConv Tone Similarity averaged 4.59, 6.17, and 4.43. Kendall's W ranged from .479 to .876 on significant axes. Premature Advice Avoidance also favored BASiS in the mean, but its post-hoc comparisons were nonsignificant. BASiS accounted for 74.4% of final choices.

In the oracle evaluation for Experimental Setting 1 shown in Figure 4, BASiS likewise had the highest mean on all seven axes. For Base, Target-only, BASiS, and Random, Style Strength averaged 8.26, 8.22, 8.65, and 8.21, and ESConv Tone Similarity averaged 8.21, 8.25, 8.52, and 8.13; BASiS significantly exceeded all three conditions on both axes. Premature Advice Avoidance averaged 8.57, 8.76, 9.44, and 8.92, again with significant differences from all three conditions. Naturalness was significantly higher than Random, while Target-only did not significantly exceed Base on any axis.

Experimental Setting 2: MathDial × WildChat-1M

Figure 5 presents the human-evaluation results for Experimental Setting 2. BASiS significantly exceeded both Base and Random on all seven axes (Holm-adjusted $p < .05$). Equitable Tutoring averaged 4.28, 5.42, and 3.73, and Feedback Actionability averaged 4.13, 5.43, and 3.72. Participants therefore favored BASiS both for preserving room for learner reasoning and for clarifying the next action. Kendall's W ranged from .716 to .939. BASiS accounted for 61.1% of final choices.

In the oracle evaluation for Experimental Setting 2 shown in Figure 6, BASiS exceeded all three comparison conditions on all seven axes; all 21 comparisons were significant (all $p < .001$). For Base, Target-only, BASiS, and Random, Equitable

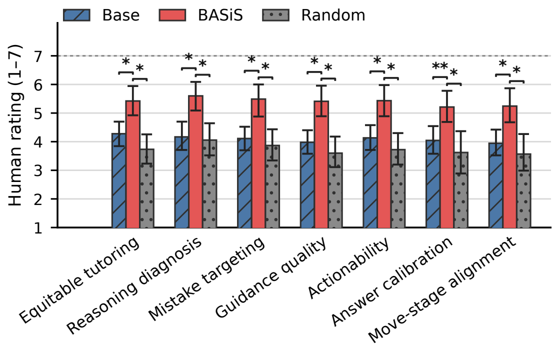

Fig. 5. Human-evaluation results for MathDial–WildChat-1M. Bars and error bars show participant-level means and bootstrap 95% confidence intervals. Brackets show significant BASiS comparisons after Holm correction ($*p < .05$, $**p < .01$).

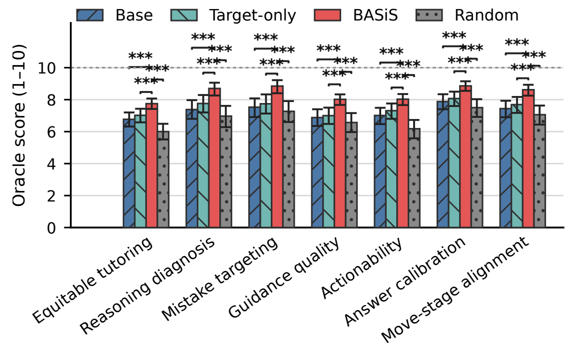

Fig. 6. Oracle-evaluation results for MathDial–WildChat-1M. Bars show condition means and bootstrap 95% confidence intervals. Brackets show significant BASiS comparisons after Holm correction over all six pairs (all $***p < .001$).

Table 5. Counts and percentages of final choices in the human evaluation

Choice	**ESConv**	**MathDial**	**MediTOD**
Base	14 (15.6%)	16 (17.8%)	11 (15.7%)
BASiS	67 (74.4%)	55 (61.1%)	31 (44.3%)
Random	6 (6.7%)	12 (13.3%)	19 (27.1%)
Almost the same	2 (2.2%)	3 (3.3%)	8 (11.4%)
Cannot judge	1 (1.1%)	4 (4.4%)	1 (1.4%)

Tutoring averaged 6.77, 7.02, 7.75, and 6.00. Learner Reasoning Diagnosis averaged 7.38, 7.76, 8.70, and 6.97, and Mistake Location and Targeting averaged 7.53, 7.74, 8.85, and 7.27. The 1.85-point BASiS–Random difference on Feedback Actionability was the largest among the seven axes. Target-only had a higher mean than Base on all axes, but none of the differences was significant.

Experimental Setting 3: MediTOD × WildChat-1M

Figure 7 presents the human-evaluation results for Experimental Setting 3. BASiS had the highest mean on all five axes. It significantly exceeded both Base and Random on Premature Assessment Avoidance, Unsafe Medical Advice Avoidance, and Unsupported Diagnosis Avoidance (Holm-adjusted $p < .05$), with Kendall's W from .799 to .813. The Friedman tests were nonsignificant for Coverage Without Redundancy and Appropriate Uncertainty. BASiS was selected most often in the final choice, at 44.3%.

In the oracle evaluation for Experimental Setting 3 shown in Figure 8, BASiS likewise had the highest mean on all seven axes. For Base, Target-only, BASiS, and Random, Response Relevance averaged 4.46, 4.70, 5.09, and 4.91, and Overall Quality averaged 5.16, 5.26, 5.69, and 5.46. Appropriate Uncertainty averaged 7.84, 7.96, 8.26, and 7.92, with BASiS significantly exceeding all three conditions. Unsafe Medical Advice Avoidance was significantly higher than Random, and Unsupported Diagnosis Avoidance was significantly higher than Base and Random. Target-only did not significantly exceed Base on any MediTOD axis.

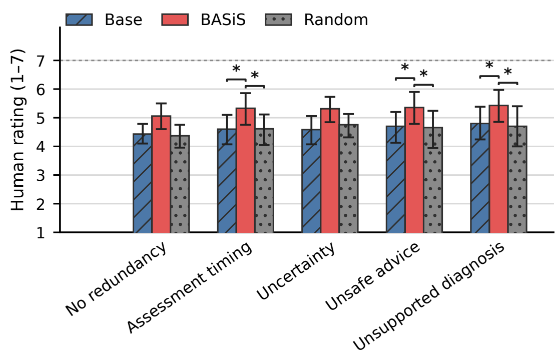

Fig. 7. Human-evaluation results for MediTOD–WildChat-1M. Bars and error bars show means for the seven complete participants and participant-level bootstrap 95% confidence intervals. Brackets show significant BASiS comparisons after Holm correction ($*p < .05$).

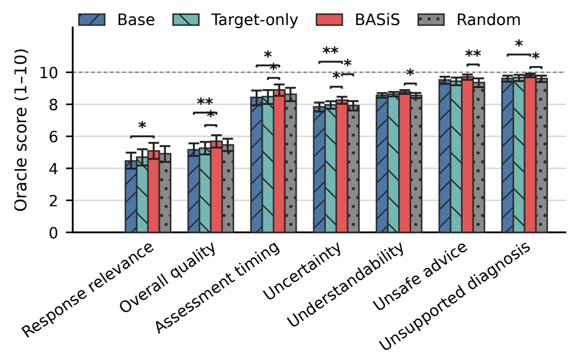

Fig. 8. Oracle-evaluation results for MediTOD–WildChat-1M. Bars show condition means and bootstrap 95% confidence intervals. Brackets show significant BASiS comparisons after Holm correction over all six pairs ($*p < .05$, $**\,p < .01$).

Table 5 summarizes the final-choice distribution. BASiS was selected most often in every domain: 74.4% for ESConv, 61.1% for MathDial, and 44.3% for MediTOD. In MediTOD, this share was not a majority but exceeded Random by 17.2 percentage points.

4.5 Discussion

We evaluated BASiS on three distinct types of purpose-specific dialogue: emotional support, mathematics tutoring, and medical history-taking. BASiS had the highest mean on all 19 human-rated axes, significantly exceeded Base and Random on 16 axes each, and was the most frequent final choice in every domain. It also had the highest mean on all 21 oracle axes and significantly exceeded Base, Target-only, and Random on 18, 13, and 15 axes, respectively. We next discuss the strengths of BASiS revealed by these results and the reasons for the observed improvements.

Discussion 1: Reproducing conversational style in emotional-support dialogue

Human raters preferred BASiS over Base and Random on style strength, an empathetic and accepting tone, supporter-role consistency, non-directiveness, contextual fit, and naturalness; BASiS also received 74.4% of final choices. Because both local expression and the broader stance of acknowledging feelings without imposing a solution were rated highly, BASiS appears to reproduce ESConv's supporter behavior as an integrated response policy. One explanation is that selecting data compatible with observation labels such as `reflect_validate` and transitions toward empathetic support teaches the model to maintain a supportive role, rather than merely increasing the use of gentle wording. Premature Advice Avoidance also had the highest human mean under BASiS, while the oracle evaluation showed significant differences from all three comparison conditions. Although the human post-hoc comparisons were nonsignificant, the mean trends agree across the two evaluations. Deferring advice therefore appears as a characteristic of BASiS that could be measured more precisely with additional dialogue contexts.

Discussion 2: Reproducing pedagogical dialogue policies in mathematics tutoring

Human raters preferred BASiS over Base and Random on all seven axes, including encouraging learner reasoning, diagnosing reasoning, locating mistakes, providing useful support, calibrating answer revelation, and matching teacher moves to the dialogue stage. BASiS also received 61.1% of final choices, indicating that participants regarded the responses as desirable mathematics tutoring overall. The oracle evaluation likewise found significant differences from all three comparison conditions on every axis, making this the most consistent pattern among the three domains. In particular, gains in Feedback Actionability indicate that BASiS did not merely generate correct content, but helped learners identify what to think about, calculate, or verify next. BASiS selects examples whose transitions from the learner's inferred state to questions, hints, or explanations match those in MathDial. This selection likely exposed the model to pedagogical scaffolding that moves a learner to the next reasoning step, explaining the broad gains from Equitable Tutoring through Teacher Move–Stage Alignment.

Discussion 3: Response quality and safety in medical history-taking

Human raters preferred BASiS over Base and Random on avoiding premature assessment, unsafe medical advice, and unsupported diagnosis. BASiS was also the most frequent final choice at 44.3%, 17.2 percentage points above Random. Its clearest strength in medical history-taking therefore appears to be a cautious response policy that withholds conclusions until sufficient information is available. By evaluating candidates through state transitions, BASiS can select responses that gather confirmation before moving to assessment, rather than responses that merely contain medically related words. Coverage Without Redundancy and Appropriate Uncertainty did not show significant human condition differences, although BASiS had the highest mean on both. The oracle evaluation also placed BASiS highest on all seven axes and found a significant advantage over all three conditions on Appropriate Uncertainty. Together, these results suggest that the benefits are clearest for safety-critical forms of restraint, while the efficiency of information gathering and the expression of uncertainty may vary more with the dialogue context. More varied symptoms and dialogue stages would allow these aspects to be examined in greater detail.

Discussion 4: Selecting dialogue progression rather than response content alone

Across all three domains, the strongest results concerned not only topic or tone, but also response timing and dialogue progression. Human raters valued acknowledging feelings in emotional support, scaffolding the learner's next step in mathematics tutoring, and avoiding premature conclusions in medical history-taking. Each policy requires the model to consider both the preceding history and the current dialogue stage. This pattern accords with Equation (3). BASiS does not score candidates solely by generic single-turn quality; it uses posterior probabilities based on state transitions and observation labels. Consequently, knowledge about *when* to offer a given kind of response is reflected in selection and appears in generated responses that humans judge desirable across the dialogue as a whole.

Discussion 5: Contributions of the target corpus and data selection

In the oracle evaluation, Target-only did not significantly exceed Base on any of the 21 axes, whereas BASiS exceeded Target-only on 13. A small target corpus can therefore be used more effectively by extracting its dialogue structure and treating that structure as a selection criterion for a large corpus, rather than using it only as direct training data. BASiS also had a higher oracle mean than Random on all 21 axes and exceeded Random on 16 of the 19 human axes. Both conditions used the same amount of training data and the same training configuration, and Random was already restricted to a coarsely matched topic category. The consistent differences therefore indicate that purpose-specific adaptation benefits from selecting examples aligned with conversational states, response strategies, and state transitions in addition to broad topical similarity.

Discussion 6: Applicability to three types of purpose-specific dialogue

We applied the same BASiS pipeline to emotional support, mathematics tutoring, and medical history-taking by changing only the reference and selection corpora. BASiS had the highest mean on every human and oracle axis and was the most frequent final choice in every domain. It can therefore extract different purpose-specific response policies rather than being limited to a single tone. The fact that the strongest gains differed by domain further illustrates the advantage of inducing states and strategies from each corpus instead of imposing one fixed guideline. Thus, although these findings provide clear initial evidence for the effectiveness of BASiS, generalization to broader user populations requires validation with external raters and more dialogue contexts. Future work should broaden the scope of evaluation and refine the design of purpose-specific states, strategies, and desirable state sets, thereby extending the strengths of BASiS to a wider range of purpose-specific dialogue applications.

5 Conclusion

We proposed BASiS (Bayesian Style-based Selection), a method that adapts a local large language model (LLM) to the conversational style required for purpose-specific dialogue through Bayesian data selection based on a small, high-quality corpus. Through constrained ontology induction, BASiS derives conversational states, response strategies, and state transitions from the small corpus and represents them as a Bayesian dialogue model. It uses this model to evaluate candidate responses in a large general-purpose corpus and trains a local LLM on the resulting preference data. This mechanism uses the response policies and multi-turn dialogue progression captured in a small corpus as criteria for selecting training data, without requiring detailed dialogue guidelines to be written manually for each purpose. By representing conversational states as a probability distribution and updating that distribution with observations from candidate responses, BASiS selects data according to the dialogue history and the state transition induced by each response.

In comparative experiments on three distinct types of purpose-specific dialogue, BASiS achieved the highest mean on all 19 human-rated axes and was the most frequent final choice in every domain. It also achieved the highest mean on all 21 oracle axes. Together, the evaluations suggest that BASiS can incorporate empathetic support, pedagogical guidance suited to learner understanding, and response policies that avoid unwarranted certainty when information is insufficient. In the oracle evaluation, Target-only did not significantly outperform Base on any axis, and BASiS broadly outperformed Random. These findings demonstrate the value of using the small corpus as a source of dialogue structure and selecting training data beyond coarse topical similarity.

Future work will evaluate other models, investigate how to design conversational states, response strategies, and desirable state sets for different purposes, and conduct human evaluations with external raters and more dialogue contexts. Through these efforts, we aim to extend BASiS to purpose-specific applications including counseling, education, and medicine.

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